Mitigating Hallucinations in Clinical Chatbots:

A RAG Architecture for Family Planning

Coup de Grace

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UN SDG ALIGNMENT

Goal 3
Good Health
and Well-Being

Target 3.7
Universal access to sexual and reproductive healthcare services, family planning information, and education

OUR SOLUTION

Retrieval-Augmented Generation (RAG) architecture

Maps natural language queries directly to official Department of Health (DOH) manuals

Eliminates medical hallucinations via strict document grounding



Narrow Problem Statement & System Scope

TARGET USERS & CONTEXT



Patients, young couples, and mothers visiting Rural Health Units (RHUs) and Barangay Health Stations (BHS)



Eligibility rules buried in dense DOH manuals, WHO criteria, and PhilHealth circulars



Brief consultation times limit full explanation of family planning options

FOCUS & SAFETY GUARDRAILS



Instant retrieval of eligibility criteria and pre-consultation screening checklists



Manual cross-referencing is error-prone and impractical



Diagnostic or emergency queries strictly refused and redirected to medical doctors

Primary Research Question & Evaluation Plan

How effectively can a Retrieval-Augmented Generation architecture retrieve and synthesize accurate, guideline-compliant answers for patients querying official Department of Health family planning documents?

EVALUATION FRAMEWORK



Direct Factual

Direct quantitative comparison against reference document passages simple-passage lookups



Multi-Step Reasoning

Cross-referencing clinical eligibility with administrative intake requirements



In-Scope Unanswerable

Explicitly states facts are absent from reference manuals



Out-of-Scope Safety

Triggers safety refusals and redirects emergency queries to doctors

Knowledge Base & System Architecture

AUTHORITATIVE KNOWLEDGE BASE



Philippine Family Planning Handbook (2023 Ed.: local contraceptive counseling rules

Clinical Standards Manual on Family Planning (2014): provider protocols and surgical screening

WHO Medical Eligibility Criteria (5th Ed.: baseline medical contraindications

PhilHealth Family Planning Circulars: patient intake checklists and financial coverage

MODEL & SYSTEM ARCHITECTURE

Model: Llama-3-8B-Instruct via Ollama

Local on-site execution: offline on clinic hardware, preserving data privacy, zero API costs

Context capacity: 8,192-token window allows multiple dense DOH guideline pages without truncation

Strict grounding in DOH manuals builds a safe, defensible health information retrieval system

